



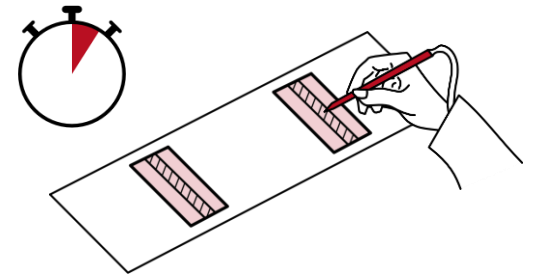
Intelligent User Interfaces

Computational UI Optimisation

Recap and Last Quiz

Lecture Quiz

Describe what **Fitts' Law** models in HCI. Also go through the provided Fitts' Law notebook. **Which pen is faster** to use for difficult tasks, based on this analysis?



$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$

Fitts' Law is a model of human movement. It allows us to compare and measure how much time it takes to point at something given the distance and the width of the object that the user intends to point at (eg a button on a touchscreen).

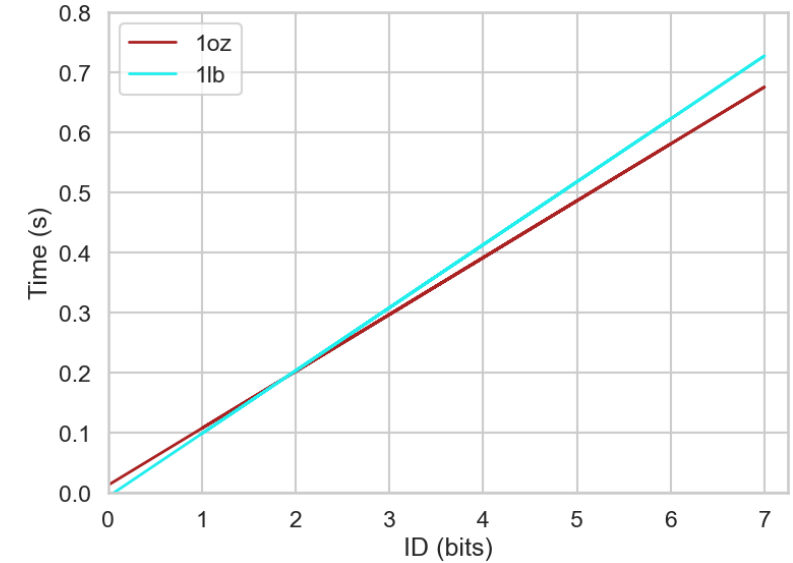
It can be expressed as an equation: $MT = a + b \log_2(2D/W)$

Intuitively, we can see that targets with a greater distance and smaller size take longer to reach while targets with shorter distance and larger size take less time to reach.

- MT = Measurement time, i.e, the time it takes for a user to reach the target.
- a, b = these are constants based on the device and the method for pointing.
- D = this refers to the distance to the target
- W = this is the size of the target.

$\log_2(2D/W)$ is also referred to as ID/Index of difficulty. We take $\log_2$ because it can help us conceptualize it in terms of information theory. So a greater ID means more bits of information, therefore the user needs to put in more precision or 'information' to hit the target.

You can read the original paper online:
<https://www.cs.princeton.edu/courses/archive/fall08/cos436/FittsJEP1954.pdf>



$$MT_{1oz} = 0.01 + 0.09 * ID$$
$$MT_{1lb} = -0.01 + 0.10 * ID$$

Lecture Quiz

You have a **new idea for an interaction device**. Briefly describe how you could **use Fitts' Law to test if it's better than a mouse**. And what does „better“ mean in this case?

Complete answer:

- **Design a user study** in which people have to click targets of varying distances (D values) and sizes (W values)
Note: Measuring only one value of D and W is not enough because we need multiple points to fit a line i.e. to compute a regression model!
- **Record the data** (D , W , MT) in this task, both for the mouse and your new device (counterbalance the order to avoid fatigue effects)
- **Compute the regression models** for both devices and compare them. „Better“ probably means lower MT at the same ID level(s).
- Practically speaking, Fitts' Law can only show something as „**better**“ in terms of **movement time**, i.e. interaction speed.

Other factors, such as errors, comfort, user experience, accessibility, etc. are not considered in a „plain“ Fitts' Law analysis!

Lecture Quiz

Think back to lecture on conversational UIs: How might a **voice assistant adapt** to an individual user? What might be the **benefit** for the user? And what **data / user model** might be needed for this?

The voice assistant could recognize specific **language patterns** of the user or learn how the user behaves in different **moods**. Based on this, the voice assistant could **phrase the answers different** and **change voice pitch and volume**.

A voice assistant could adapt by learning a user's **habits**, vocabulary, accent, and **preferred response style**. For example, if a user usually wants quick answers, the assistant could become more direct. If the user often says "study mode", it could learn that this means turning off notifications, starting focus music, and opening notes.

Data for adaptations:

- User profile such as Name, age, gender, language, preferred voice settings.
- Interaction history
- Learned pattern of user's speech
- User preferences such as favorite apps, services, contacts, playlist and content

This lecture: Learning goals

After this session, you should be able to:

- ✓ **Explain** the general idea and motivation of using optimisation as a UI design approach
- ✓ **Explain** its key components, with examples
- ✓ **Analyse** a given UI to give a recommendation of how we might formalize its properties for computational optimisation
- ✓ **Implement** Simulated Annealing for UI optimisation
(in pseudo-code and/or a language you're familiar with)

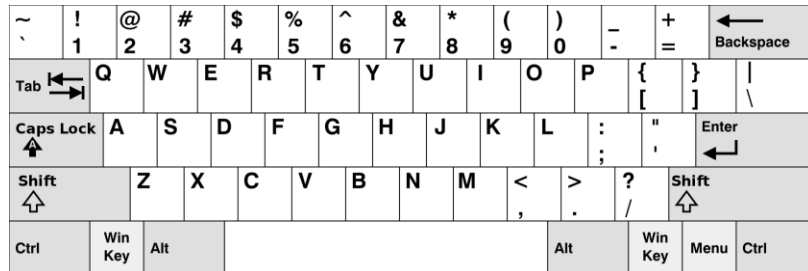


Intro and motivation

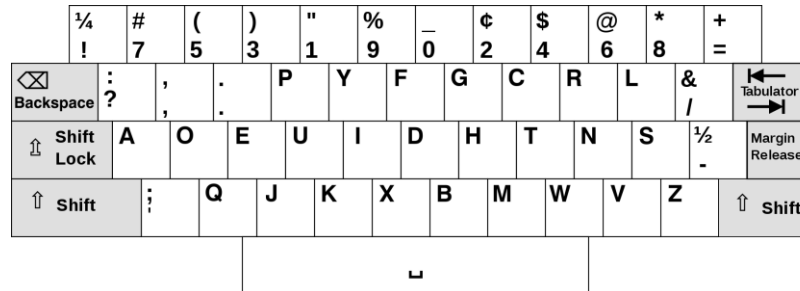
Our keyboard example

Keyboards as a challenging UI layout problem

- Are some layouts better than others?
- If so, how do we find the best one?



QWERTY, by Christopher Sholes, 1873



Dvorak, by August Dvorak, 1936



By: https://commons.wikimedia.org/wiki/File:KB_United_States.svg, https://commons.wikimedia.org/wiki/File:KB_United_States_Dvorak.svg

UI element assignment problem

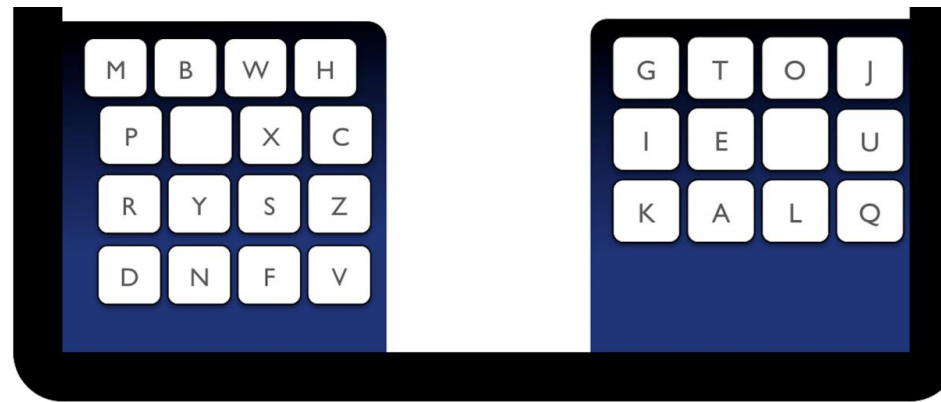
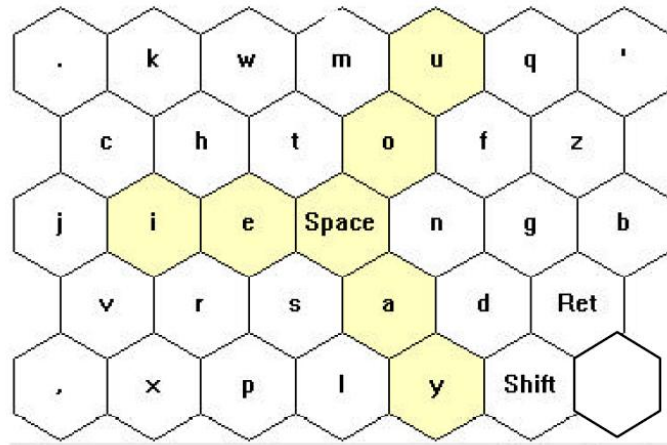
Here: key assignments



How many layouts are there? $26! = 4 * 10^{26}$

For comparison - stars in the universe: https://www.esa.int/Science_Exploration/Space_Science/Herschel/How_many_stars_are_there_in_the_Universe

Examples of optimised designs

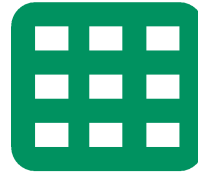


[Zhai et al. 2000, Dunlop and Levine 2012, Oulasvirta et al. 2013, Gong et al. 2018]

Design as optimisation

What is “optimal”?

- **Design space:**
Best among which options?
- **Design objective:**
Best for what?
- **Optimizer:**
How to find the best design?
Best with which guarantees?



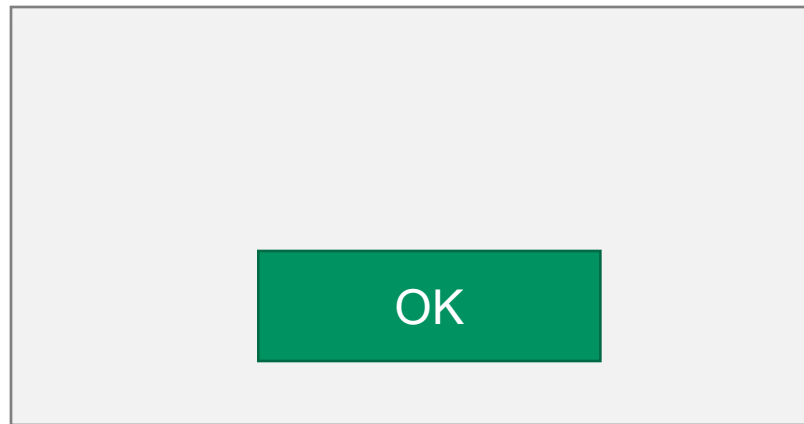
Design space, formalised

$$d = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix}$$

Design space ***D***
with ***n*** design variables

Q: Which design variables might we consider in this simple UI?

e.g. button x, y, width, height



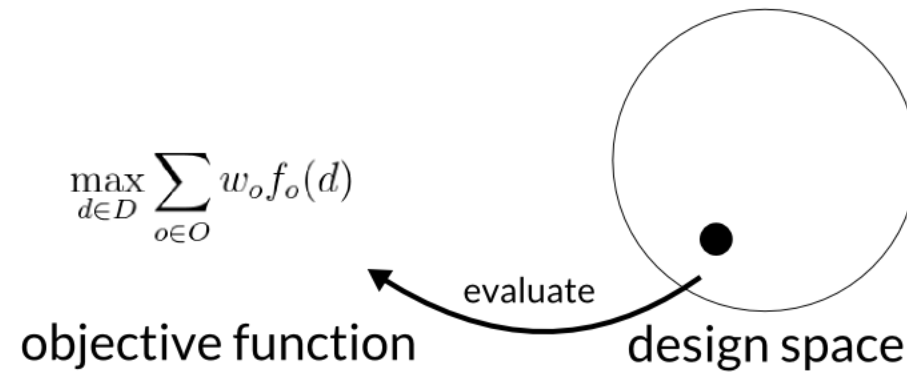
Design space for the key assignment

= Set of possible layouts



Objective function

Here: How to judge a keyboard layout?



How to judge a keyboard layout?

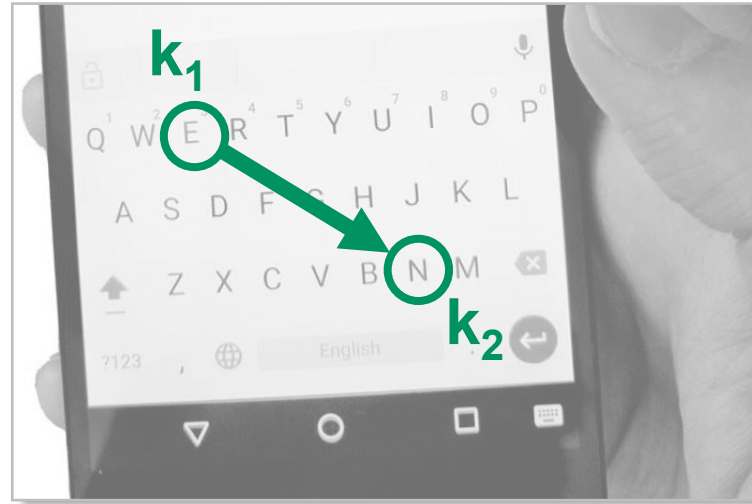
Also see previous lectures for pointing and language models

- Finger movement time
(e.g. Fitts' law)

$$t(k_1, k_2) = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

- Language properties
(e.g. bigram frequencies)

e.g. $p("n"|"e") = 0.001$



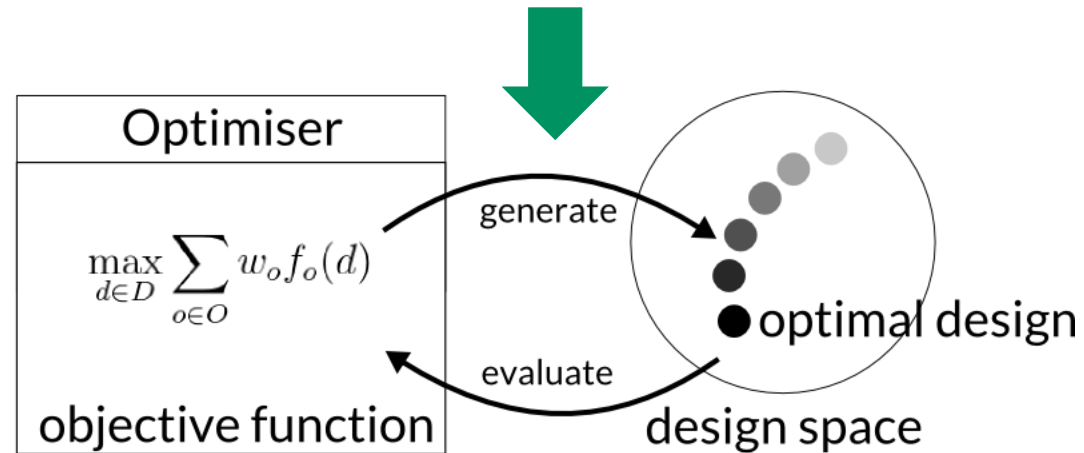
- Combined: mean time between two key presses

$$f(d) = \sum_{k_1 \in K} \sum_{k_2 \in K} p(d(k_2)|d(k_1)) t(k_1, k_2)$$

where the design d maps from keys to characters

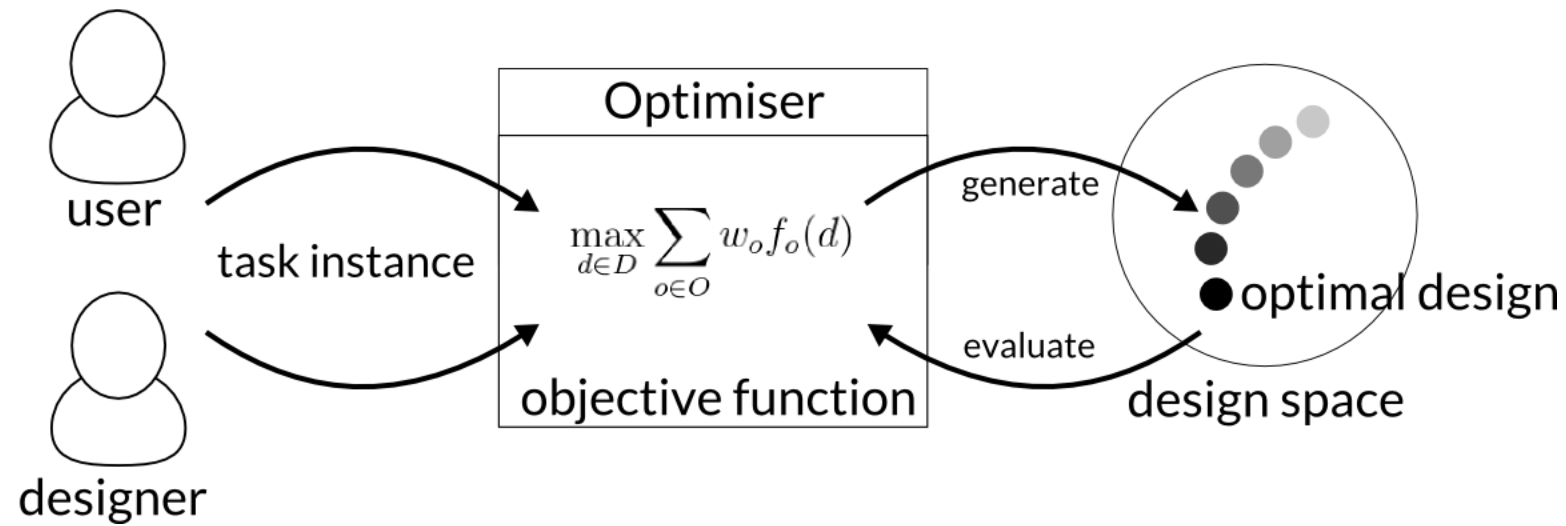
Optimiser

Here: How to pick layouts?



Design task

e.g. keyboard layout optimisation



Optimisation methods

For UI design

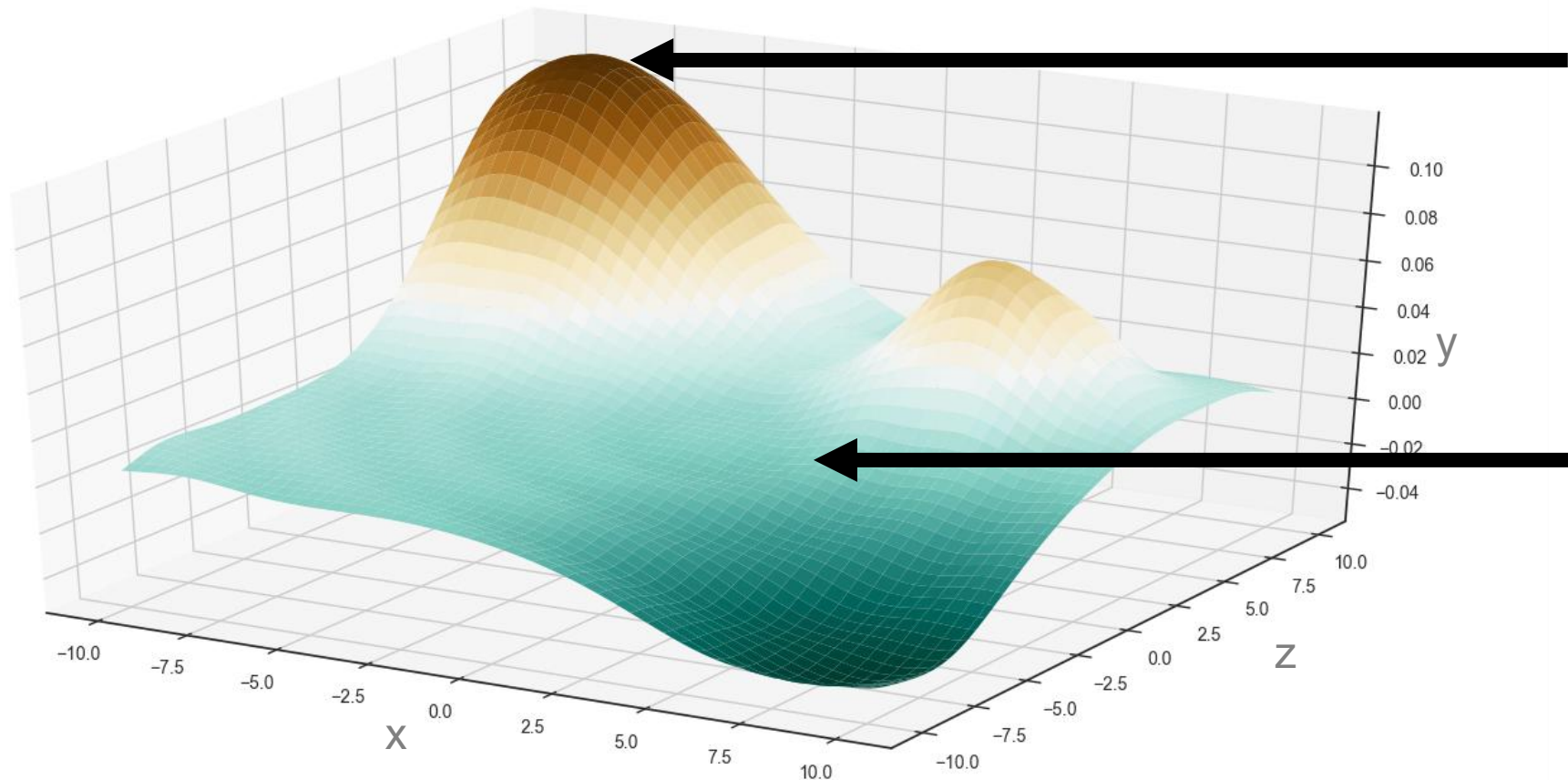
A simple optimiser

- Can you think of a trivial optimiser?

- Random Search:
 1. Generate random design
 2. Evaluate it with the objective function
 3. Keep it if it's better than the current best design
 4. Repeat

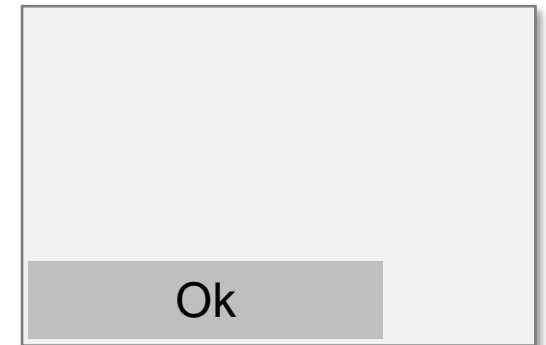
Optimisation landscape

Here: objective function (y) across two design parameters (x, z)

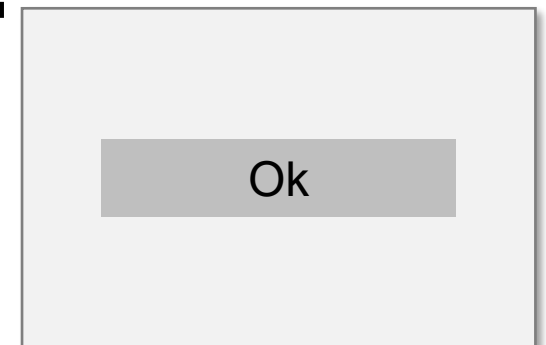


Example:
2 design parameters:
location of a button
(but could also be size, colour, etc.)

Button at -10,10
→ high objective value

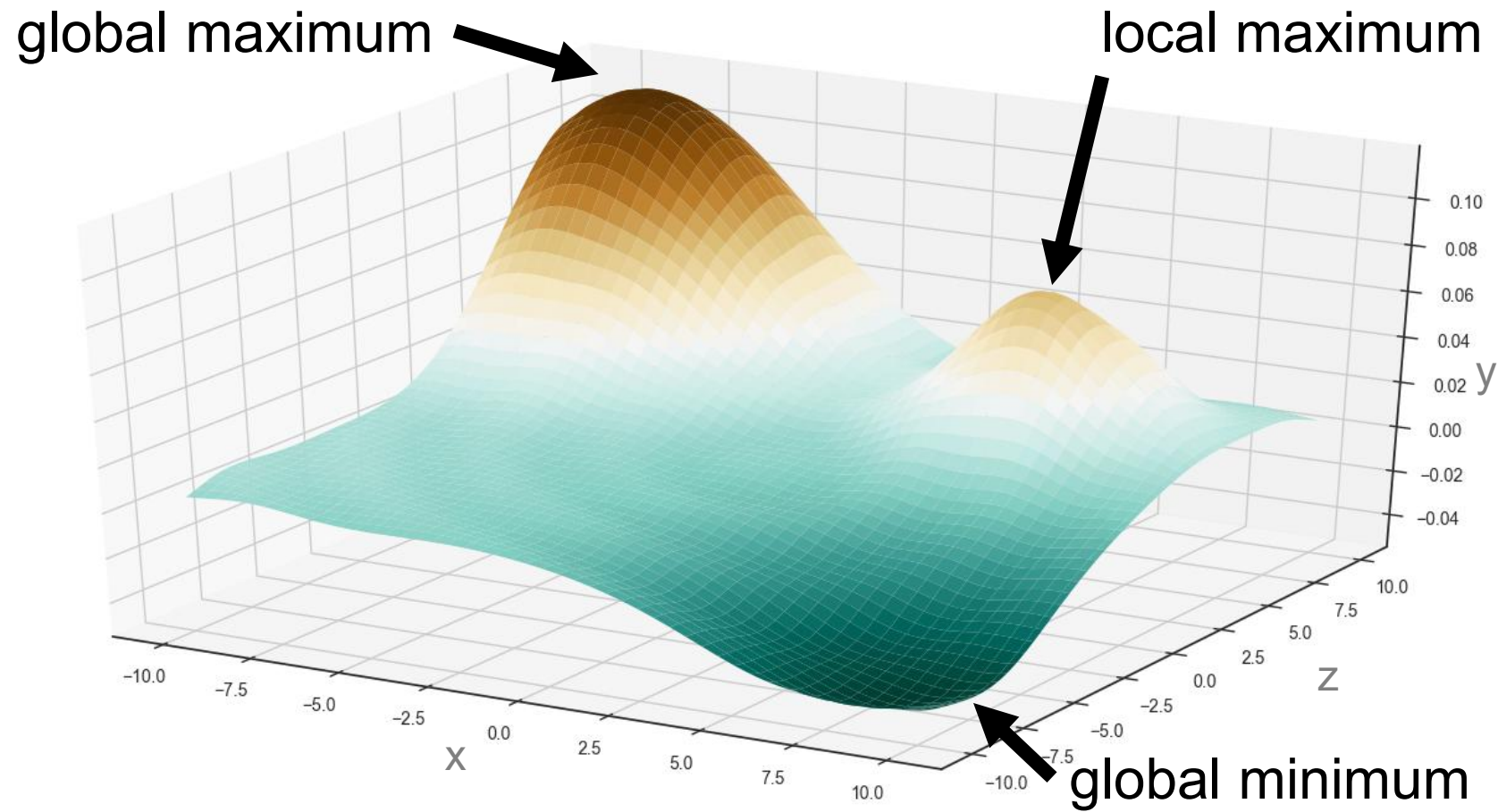


Button at 0,0
→ low objective value

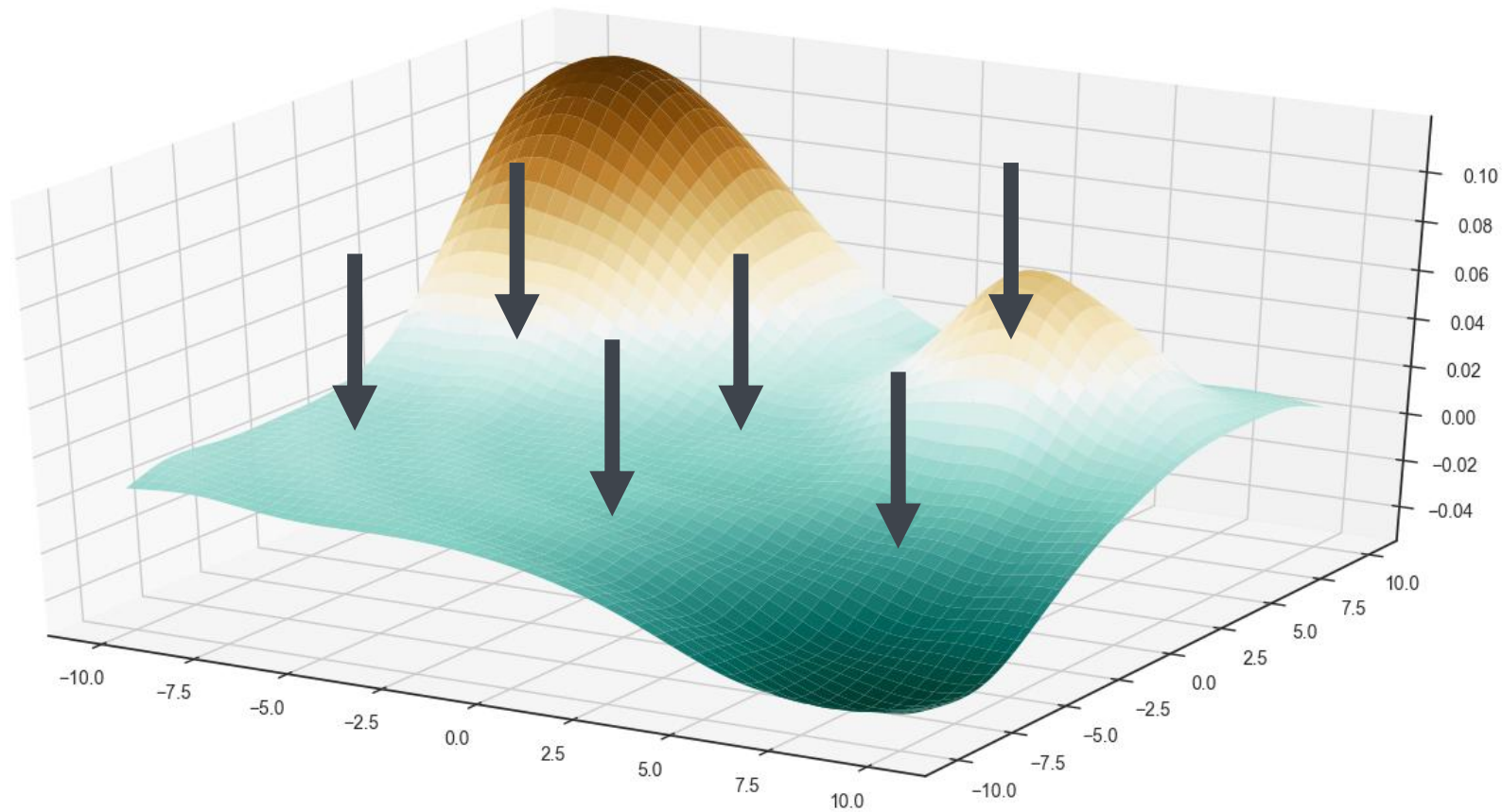


Optimisation landscape

Here: objective function (y) across two design parameters (x, z)



Random guessing



Optimisers

A very short overview

- **Heuristic methods** (e.g. Simulated Annealing)
 - + Flexible
 - Not guaranteed to find global optimum
- **Exact methods** (e.g. Integer Programming)
 - + Guarantees
 - Less flexible objectives

If you're interested in Integer Programming for this problem, see: <https://dl.acm.org/doi/10.1145/2642918.2647382>

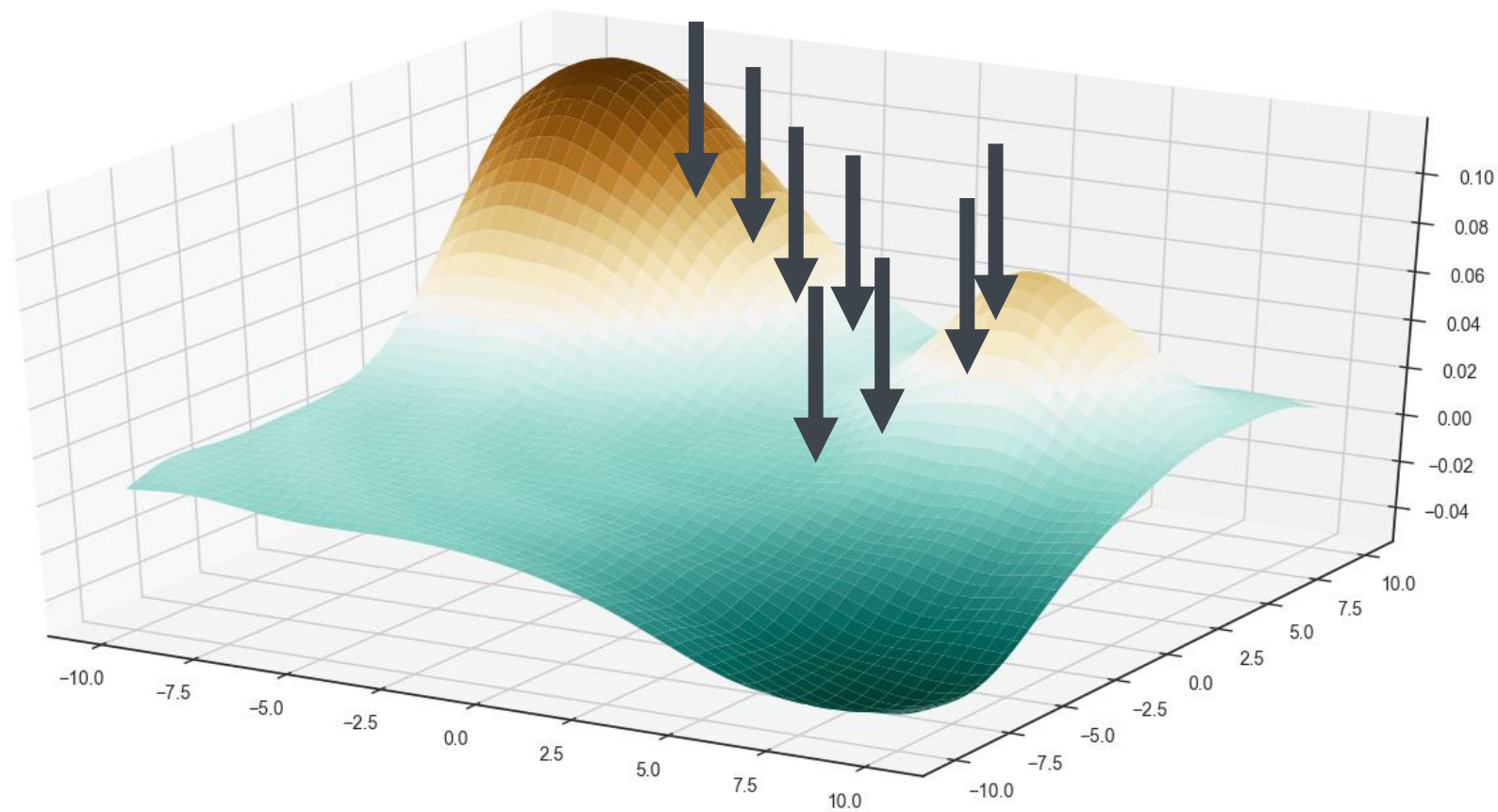
Example: Simulated Annealing

- Metaphor: shaping hot metal
- Flexible at beginning (exploration)
- Gradually more rigid as it “cools down” (exploitation)

```
for i=0 to N:  
    reduce temperature T  
    generate neighbor design  
    if better:  
        go to neighbor  
    else:  
        still go with chance relative to T
```



Simulated Annealing



Example results

“boxpum”

found with random search

WPM: 31.97

j k z f b o x p u m

d n i t c y r l v

h a e s g q w

“aero”

found with Simulated Annealing

WPM: 36.61

x y c h t i n d k z

b l a e r o f g j

v p m s u w q

Challenge – can you find a better layout than “aero”?
Use the provided python notebook as a starting point.

Example Results

With a modified objective function

“chat”

found with Simulated Annealing

WPM: 34.56

z x y v p l d u s m

q j b f r e o n i

k w g c h a t

Q: What was this layout optimised for?

→ Typing with the right thumb, reduce thumb stretching

Activity

Optimising UIs beyond keyboards

- What other (parts of) UIs might be interesting to optimise with this approach?
 - What would you have to adapt from the keyboard case?
 - What might be challenging?
-
- Ca. 3 minutes

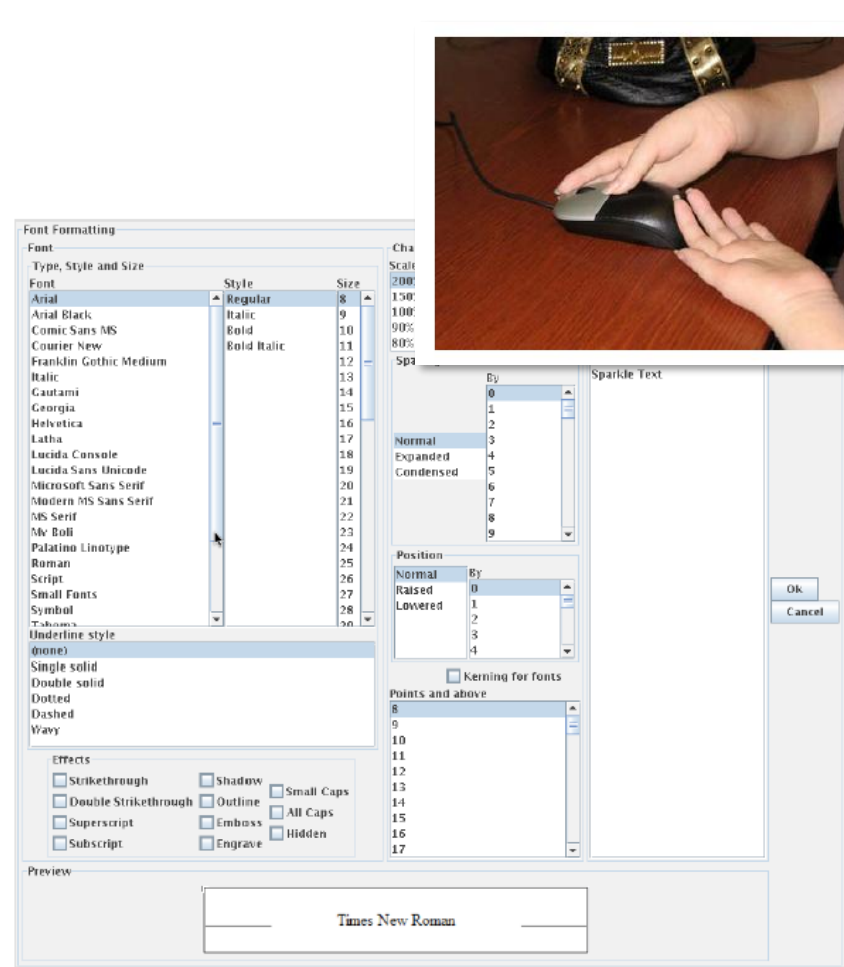
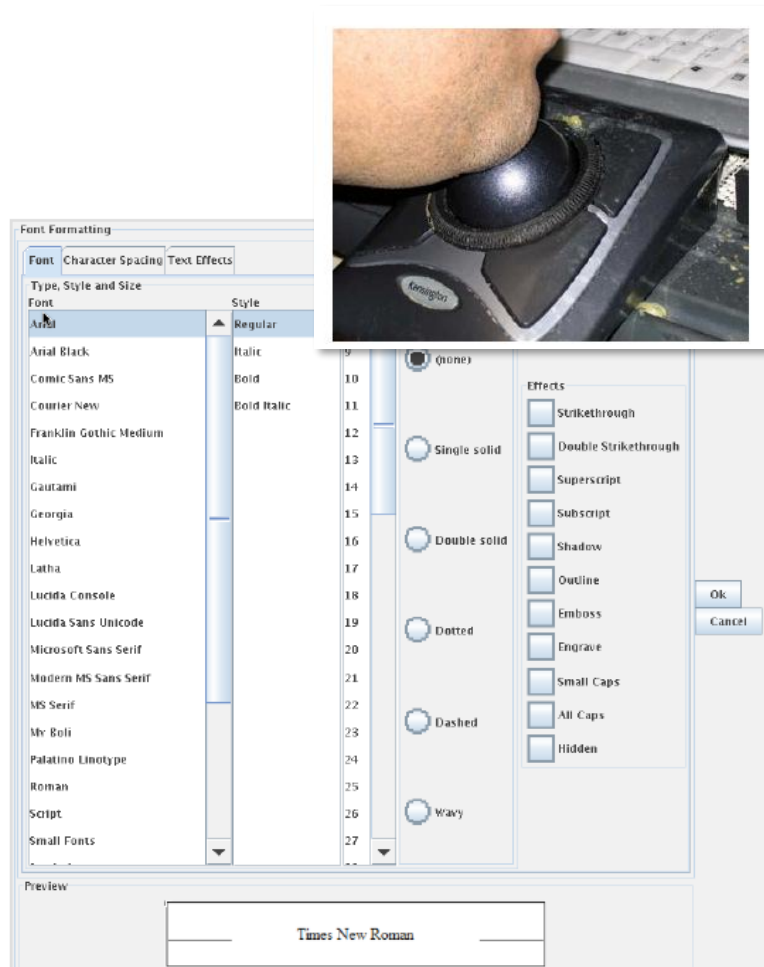


Further examples

Going beyond keyboards

Remember the user modelling lecture?

Ability-based optimisation

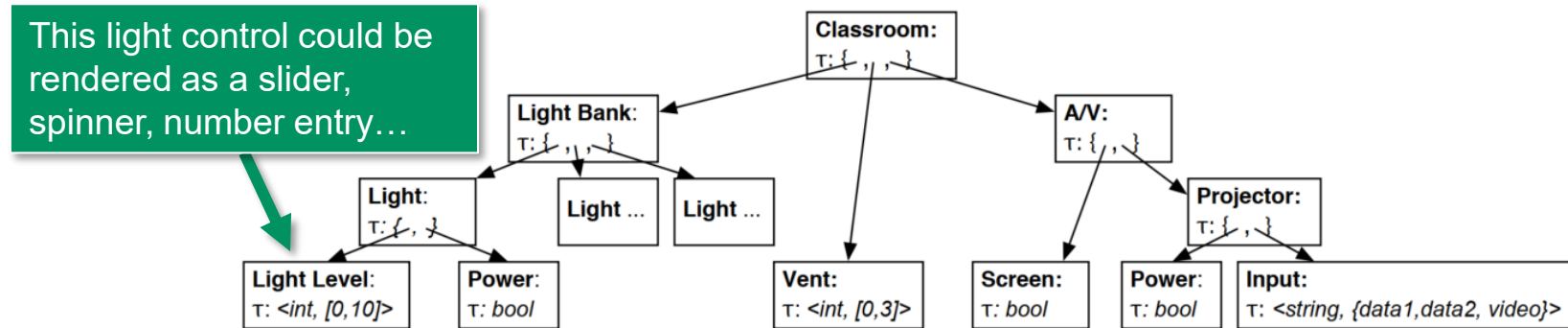


[Gajos et al., 2008]

Example: Ability-based optimisation

How does it work? → Design as search [Gajos et al., 2004 & 2008]

- Functional specification tree defines a **search space**



- Cost function** estimates interaction time

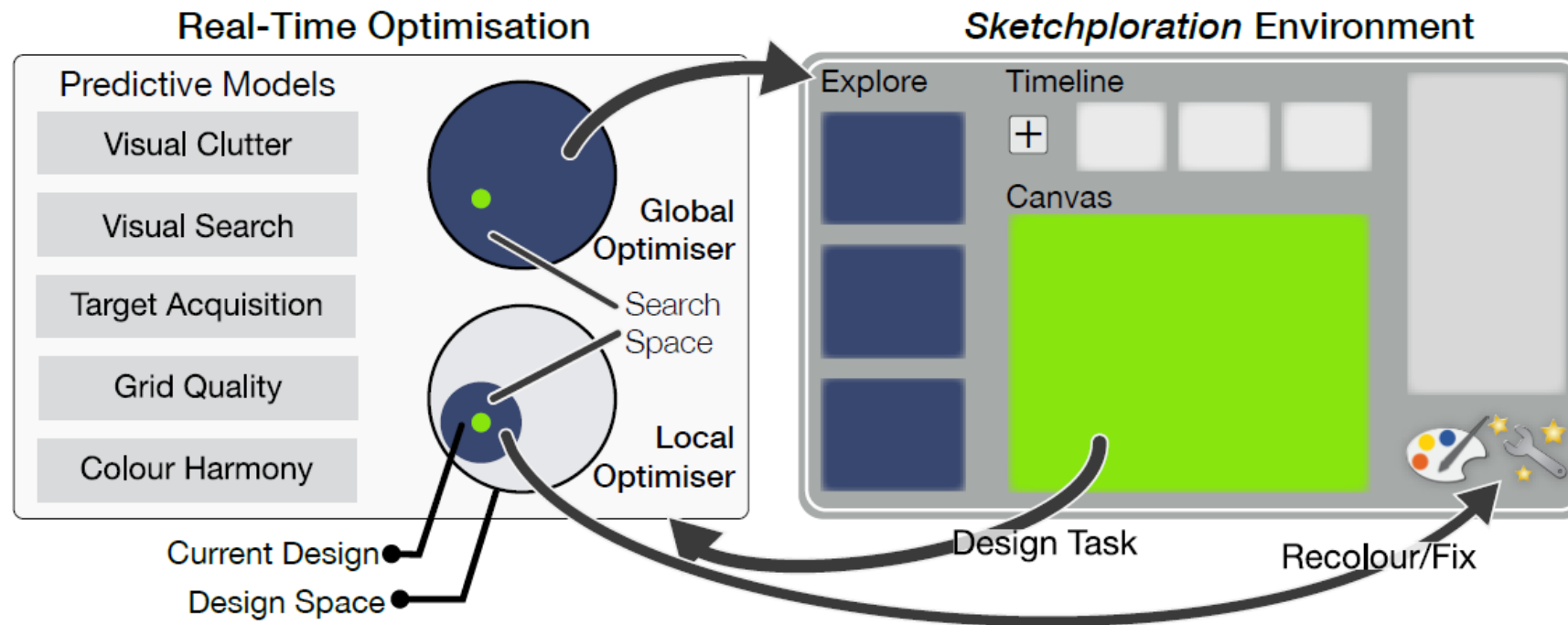
$EMT(R(S_f, s), T)$ Cost here: User's **E**stimated **M**ovement **T**ime in a **R**endered UI for specification S_f with minimum target size s , and user „trace“ T (user's typical action sequence)

→ **A combinatorial optimisation / search problem:**

Use an optimisation / search algorithm to find the UI choices for the specified functions that minimise the interaction cost

Example: „Sketchplore“

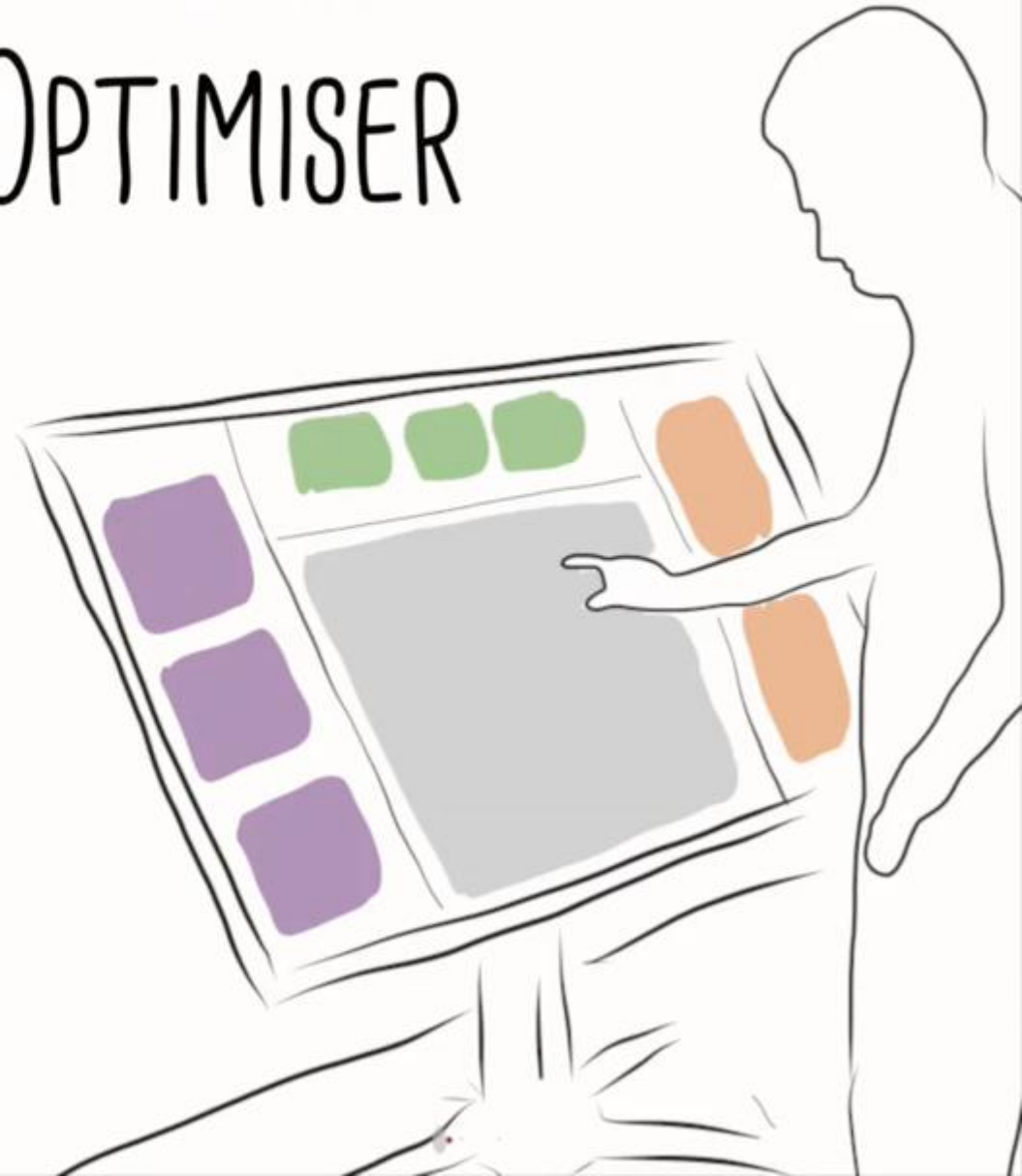
An interactive GUI layout optimiser



[Todi et al., 2016]

SKETCHPLORE: SKETCH AND EXPLORE LAYOUT DESIGNS WITH AN OPTIMISER

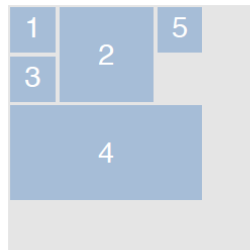
KASHYAP TODI
DARYL WEIR
ANTTI OULASVIRTA



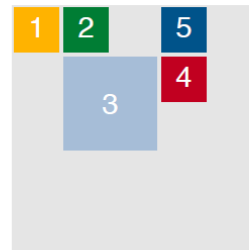
Example: „Sketchplore“

An interactive GUI layout optimiser

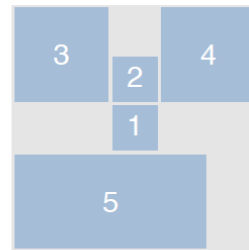
Integrate multiple models:



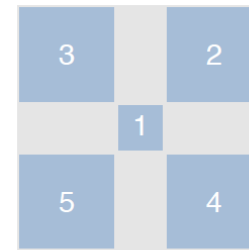
(a) Clutter



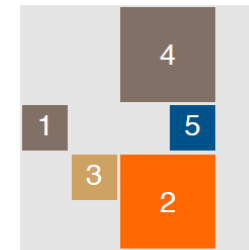
(b) Visual search



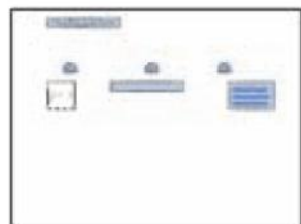
(c) Target acquisition



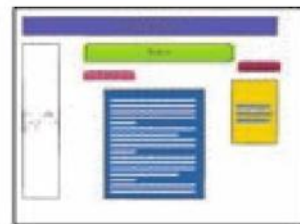
(d) Grid quality



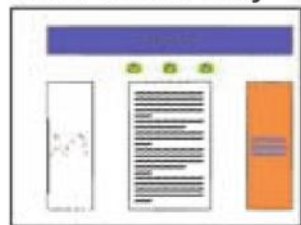
(e) Colour harmony



Color clutter only



Visual search only



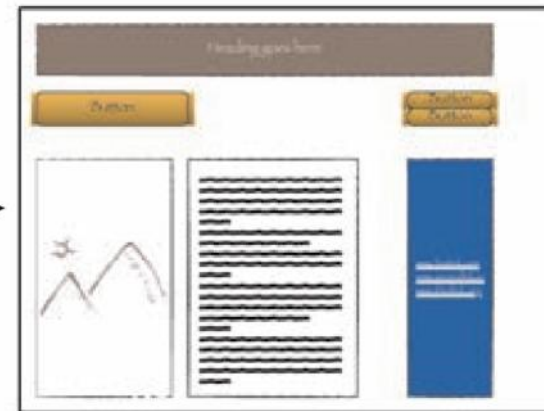
Grid quality only



Selection time only



Color harmony only

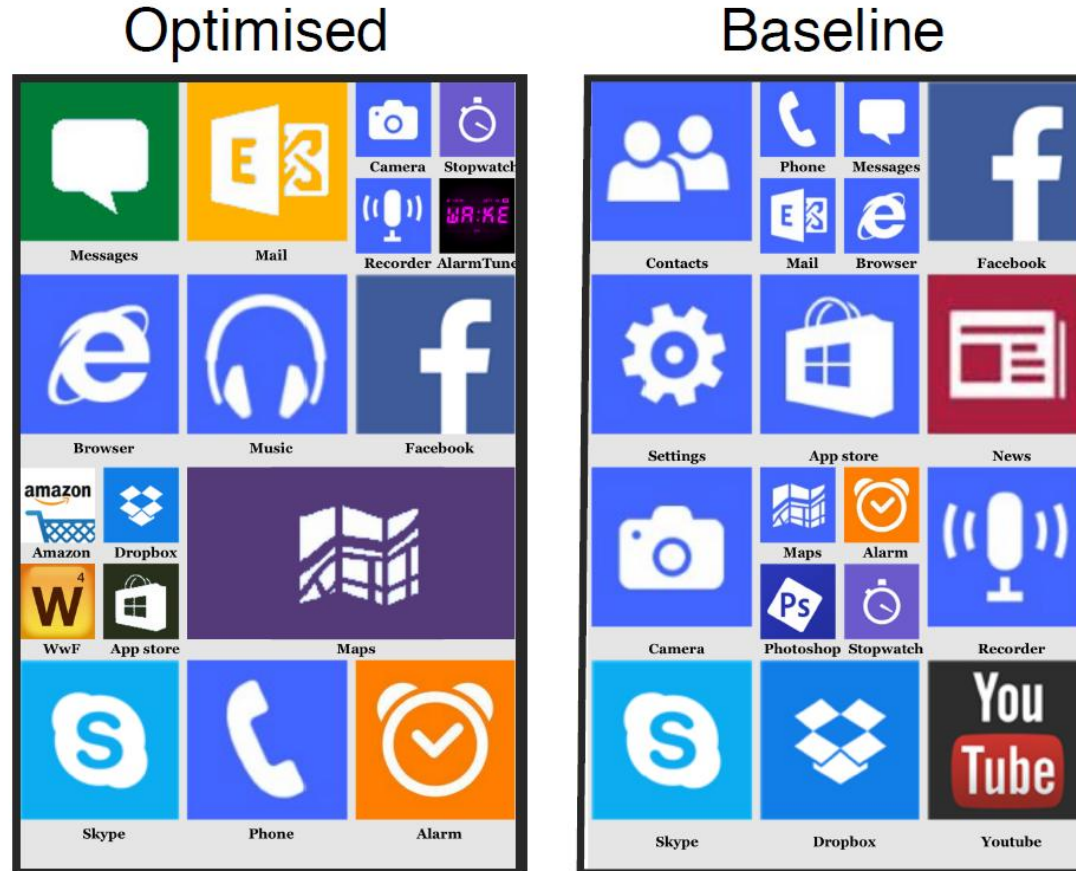


All objectives

[Todi et al., 2016, Oulasvirta 2017]

Example: „Sketchplore“

An interactive GUI layout optimiser

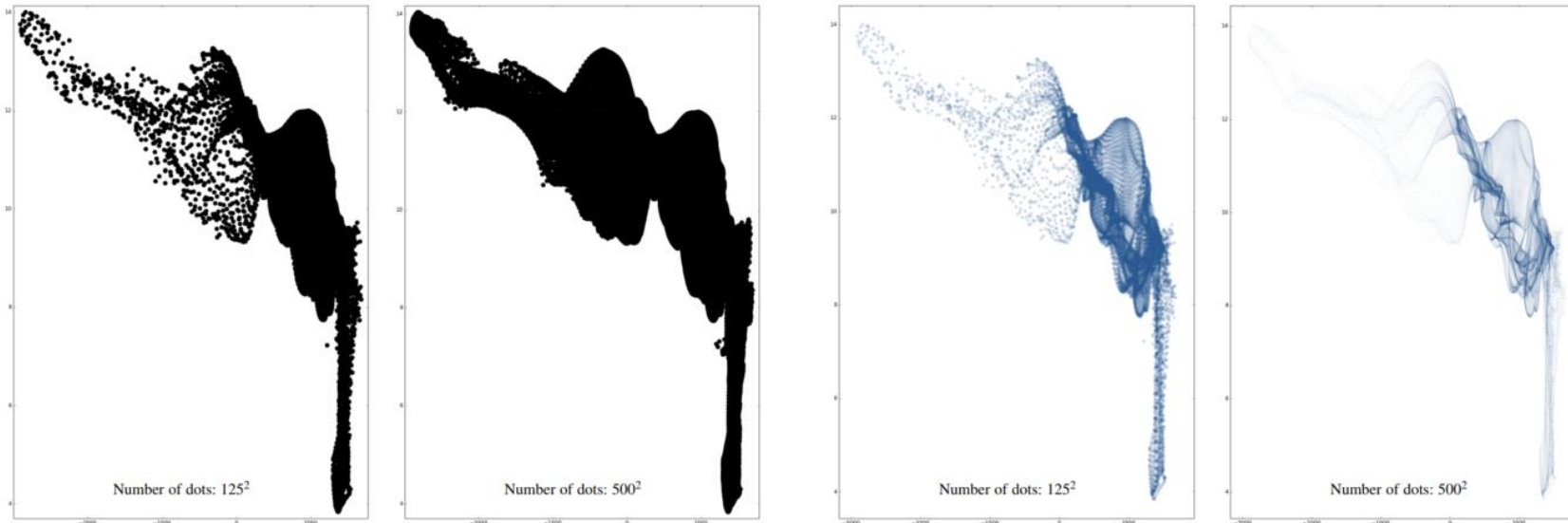


[Todi et al., 2016]

Example: Scatterplot optimisation

Motivation and idea

- The same data can be plotted in many different ways



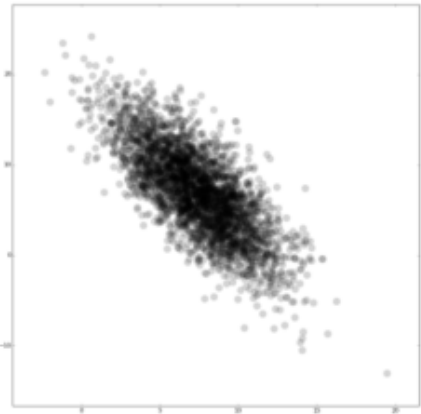
- Idea: Optimise image parameters to best support viewer in certain tasks (e.g. estimate correlation, spot outliers)

[Micallef et al., 2017]

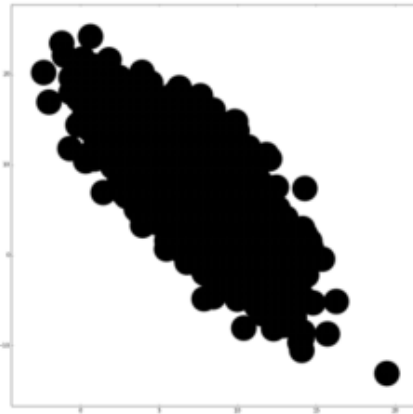
Example: Scatterplot optimisation

Optimised design parameters

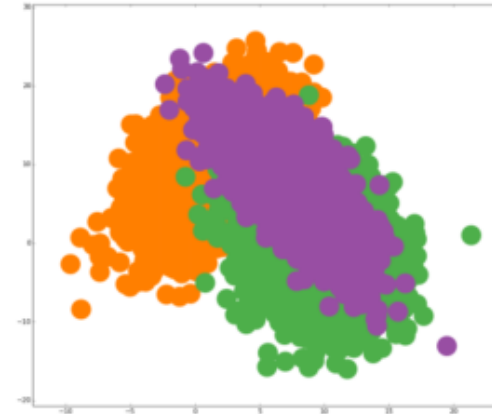
- Marker size
- Marker opacity
- Aspect ratio of the whole plot



Optimised for
perceiving correlation



Optimised for
spotting outliers



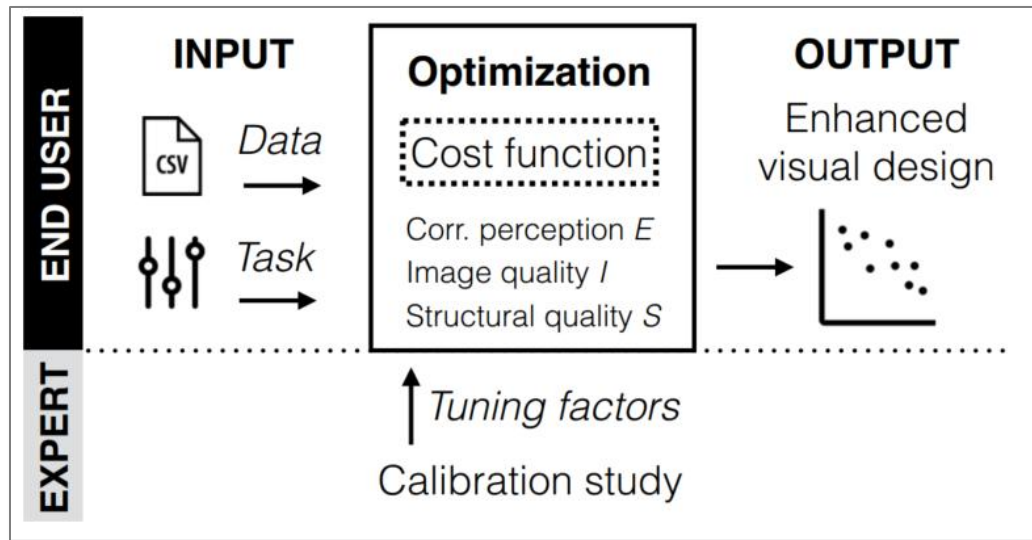
Optimised for
seeing groups

[Micallef et al., 2017]

Example: Scatterplot optimisation

Approach and components

- Approach: Optimisation with perceptual measures



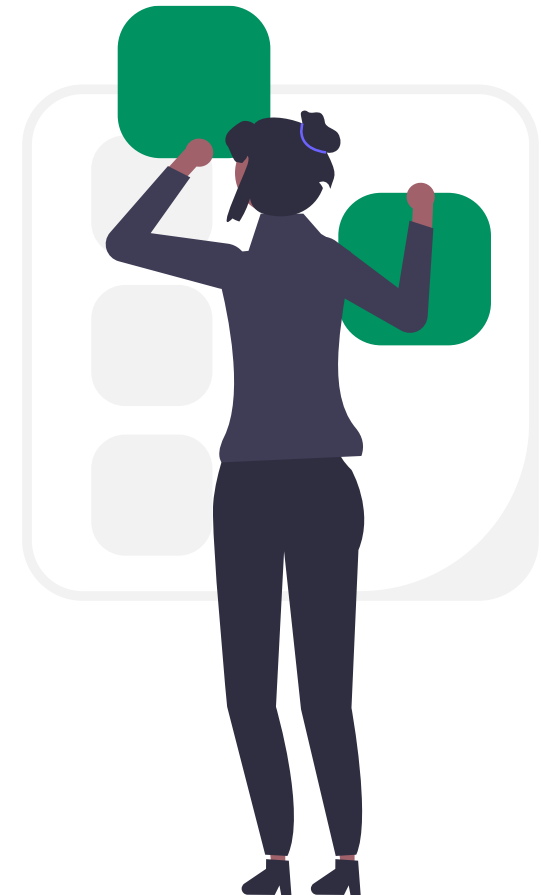
- Components of the cost function:
 - Image quality (e.g. contrast)
 - Perception of correlations in the visualised data
 - Perception of outliers
 - Perception of classes

[Micallef et al., 2017]

Summary and Outlook

Potential and challenges

- Improving quality and robustness of designs
- Exploring a large design space comprehensively
- Obtaining information on the design problem and a formal specification
- Estimating possible improvements
- Supporting human designers
- ...
- **But requires:**
Models of user behaviour, formal problem definition / objective function, computational capacity, ...



Reflections and limitations

Potential downsides of optimisation as a design method?

- E.g. bias towards caring about easily quantifiable aspects (but less about experience / broader aspects)
- Innovation vs optimisation?
“building the right thing vs building the thing right”

Other questions:

- Should all designers be developers?
- How to integrate these methods into design practice and tools?

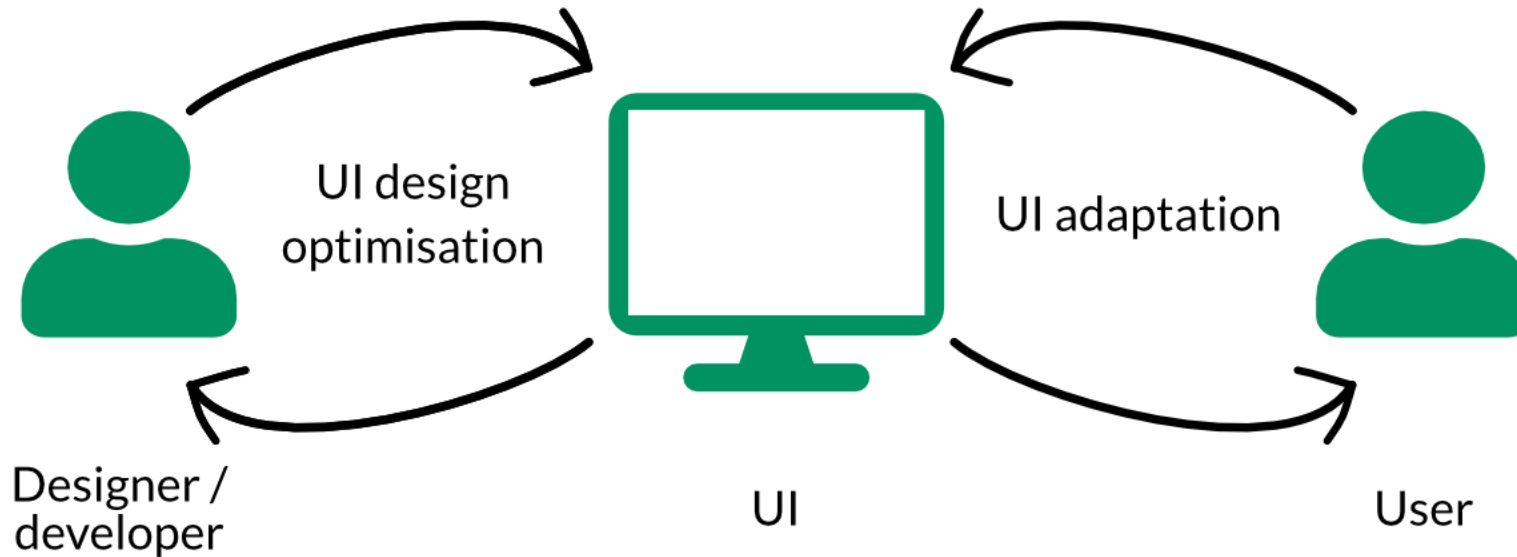
Comparison to Machine Learning

- A different way than ML of utilising computational power to improve UIs
- In particular: Can work without (user-specific) data
- E.g. by using existing models instead
- Also from other fields
(see model collection at the end of lecture 2)



Model	Example objective	Reference
Fitts' law	Minimize finger travel times	Zhai et al. UIST 2000
Search-Decide-Point	Minimize visual search and selection time in menus	Bailly et al. UIST 2013
Coherence	Ensure related items are close to each other	Bailly et al. UIST 2013
Familiarity	Ensure spatial resemblance with a prior design	Dunlop et al. CHI 2013
Keystroke-level model	Minimize task completion time for a sequential user task	Sancar et al. IsAP 2016
Active vision	Minimize visual search time when items are colored	Todi et al. DIS 2016
Rosenholtz clutter model	Minimize visual clutter, maximize saliency	Rosenholtz et al. 2011
Color harmony theory	Ensure harmonious color sets	Todi et al. DIS 2016
Motor complexity theory	Minimize complexity of gesture sets	Sridhar et al. CHI 2015
Ergonomics	Minimize shoulder stress	Bachynskyi et al. TOCHI 2015
Internal clock model	Keep temporal task performance at desirable level	Lee et al. CHI 2016

Comparison to UI adaptation



Design as optimisation:

At design time, before use

UI adaptation:

At usage time, during or after use

→ But also overlaps:

E.g. use optimisation methods to achieve adaptation (see ability-based UIs example)

Questions & Discussion



- Name and explain the key components of optimisation-based UI design.
- How can designers influence obtained designs in this approach?
- Explain Simulated Annealing. Can you consider a design resulting from this method as „optimal“? Why (not)?
- If it is possible to find better designs than QWERTY, why are we not using them widely?
- Beyond keyboard layouts, which other UI design problems could be addressed with this approach? And which are hard to address in this way?
- Create your own „exam question“ (+ answer) for this lecture:

Material

- If you're interested in more background on the overall perspective of “design as optimisation”, you might enjoy reading ***Simon, H. A. The sciences of the artificial.***

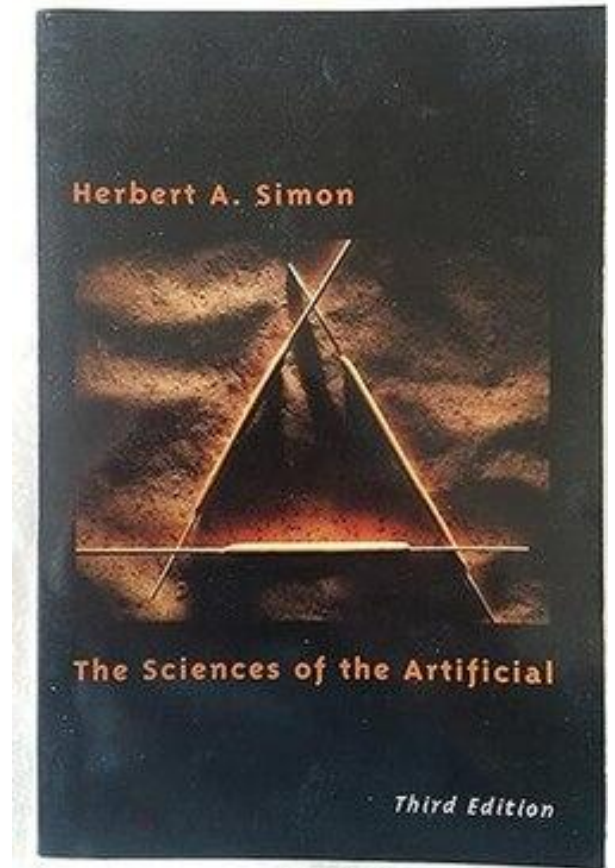


Image from: https://en.wikipedia.org/wiki/The_Sciences_of_the_Artificial

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- Oulasvirta, A., Reichel, A., Li, W., Zhang, Y., Bachynskyi, M., Vertanen, K., & Kristensson, P. O. (2013). Improving two-thumb text entry on touchscreen devices. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2765–2774. <https://doi.org/10.1145/2470654.2481383>

References (2)

- Todi, K., Weir, D., & Oulasvirta, A. (2016). Sketchplore: Sketch and Explore with a Layout Optimiser. *Proceedings of the 2016 ACM Conference on Designing Interactive Systems*, 543–555. <https://doi.org/10.1145/2901790.2901817>
- Zhai, S., Hunter, M., & Smith, B. A. (2000). The metropolis keyboard—An exploration of quantitative techniques for virtual keyboard design. *Proceedings of the 13th Annual ACM Symposium on User Interface Software and Technology - UIST '00*, 119–128. <https://doi.org/10.1145/354401.354424>

Further Reading:

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